

Topic 3: Classification and Biodiversity

This topic considers the evidence used in the development of models for the classification of organisms. It also considers the limitations of these models. The topic includes the principles underlying natural selection and how this can lead to speciation.

Opportunities for developing mathematical skills within this topic include the principles of sampling as applied to scientific data, and assessment of species diversity using a formula to calculate an index of diversity. (Please see *Appendix 6: Mathematical skills and exemplifications* for further information.)

Students should:

3.1 Classification

- i Know that the classification system consists of a hierarchy of domain, kingdom, phylum, class, order, family, genus and species.
- ii Understand the limitations of the definition of a species as a group of organisms with similar characteristics that interbreed to produce fertile offspring.
- iii Understand why it is often difficult to assign organisms to any one species or to identify new species.
- iv Understand how gel electrophoresis can be used to distinguish between species and determine evolutionary relationships.
- v Know that DNA sequencing and bioinformatics can be used to distinguish between species and determine evolutionary relationships.
- vi Understand the role of scientific journals, the peer review process and scientific conferences in validating new evidence supporting the accepted scientific theory of evolution.
- vii Understand the evidence for the three-domain model of classification as an alternative to the five-kingdom model and the role of the scientific community in validating this evidence.